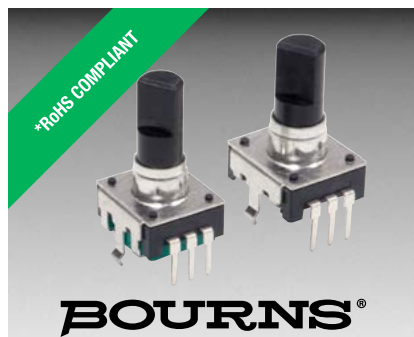




Features

- Compact design, long life and high reliability
- Momentary push switch option
- Available in a wide variety of configurations to meet many user requirements



PEC12R - 12 mm Incremental Encoder

Electrical Characteristics

Output.....	2-bit quadrature code
Contact Rating.....	10 mA @ 5 VDC
Insulation Resistance.....	10 megohms @ 250 VDC
Dielectric Withstanding Voltage Sea Level.....	50 VAC minimum
Electrical Travel.....	Continuous
Contact Bounce (15 RPM).....	2.0 ms. maximum**
RPM (Operating).....	100 maximum**

Environmental Characteristics

Operating Temperature Range.....	-30 °C to +70 °C (-22 °F to +158 °F)
Storage Temperature Range.....	-40 °C to +85 °C (-40 °F to +185 °F)
Humidity.....	MIL-STD-202, Method 103B, Condition B
Vibration.....	10~55~10 Hz / 1 min. / Amplitude 1.5 mm
Shock.....	100 G
Rotational Life.....	30,000 cycles minimum
Switch Life.....	20,000 cycles minimum
IP Rating.....	IP 40
Moisture Sensitivity Level.....	1
ESD Classification (HBM).....	N/A

Mechanical Characteristics

Mechanical Angle.....	360 ° continuous
Torque	
Detent.....	30 to 90 gf-cm (0.41 to 1.25 oz-in)
Running.....	10 to 70 gf-cm (0.14 to 0.97 oz-in)
Mounting.....	10.2 kgf.cm (8.83 lb.-in.) maximum
Shaft Side Load (Static).....	2.04 kgf (4.5 lbs.) minimum
Weight.....	3 gm (0.1 oz.) maximum
Terminals.....	Printed circuit board terminals
Terminals.....	Printed circuit board terminals
Soldering Condition	
Wave Soldering.....	Sn95.5/Ag2.8/Cu0.7 solder with no-clean flux: 260 °C max. for 3 ± 1 sec.
Hand Soldering.....	Not recommended
Hardware.....	One flat washer and one mounting nut supplied with each encoder with bushing

Switch Characteristics

Switch Type.....	Contact Push ON Momentary SPST
Power Rating (Resistive Load).....	10 mA at 5 V DC
Switch Travel.....	0.5 ± 0.3 mm
Switch Actuation Force.....	610 ± 306 gf (8.47 ± 4.24 oz.-in.)
Contact Resistance.....	100 milliohms @ 5 VDC

How To Order

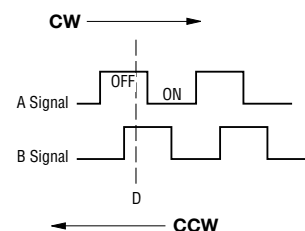
Model	PEC12R - 4 0 20 F - S 0012
Terminal/Bushing Configuration	
2 = Vertical Mount - Radial PC Pin/No Bushing	
3 = Horizontal Mount - Axial PC Pin/with Bushing	
4 = Horizontal Mount - Axial PC Pin/No Bushing	
Detent Option	
0 = No Detents 1 = 12 Detents (available with 12 pulses only) 2 = 24 Detents	
Standard Shaft Length	
15 = 15.0 mm (Horizontal Mount/No Bushing only)	22 = 22.5 mm
17 = 17.5 mm	25 = 25.0 mm
20 = 20.0 mm	30 = 30.0 mm
Shaft Style	
F = Insulated Flatted Shaft	
Switch Configuration	
S = Push Momentary Switch N = No Switch	
Resolution	
0012 = 12 Pulses per 360 ° Rotation 0024 = 24 Pulses per 360 ° Rotation	

Additional Information

Click these links for more information:



Quadrature Output Table



BOURNS®

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Email: eurocus@bourns.com

The Americas:

Tel: +1-951 781-5500

Email: americus@bourns.com

www.bourns.com



WARNING

Cancer and Reproductive Harm

www.P65Warnings.ca.gov

*RoHS Directive 2015/863, Mar 31, 2015 and Annex.

**Devices are tested using standard noise reduction filters. For optimum performance, designers should use noise reduction filters in their circuits. Specifications are subject to change without notice. Users should verify actual device performance in their specific applications. The products described herein and this document are subject to specific legal disclaimers as set forth on the last page of this document, and at www.bourns.com/docs/legal/disclaimer.pdf.



CAUTION

Do not store product in high temperature and humidity, direct sunlight and/or places where corrosive gases may be generated. Please use product within 6 months from the date of delivery and promptly after unpacking.

Applications

Level control, tuning and timer settings in:

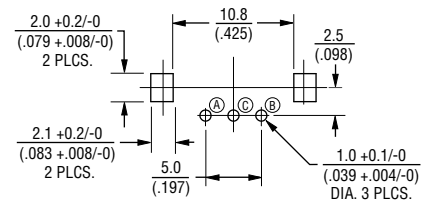
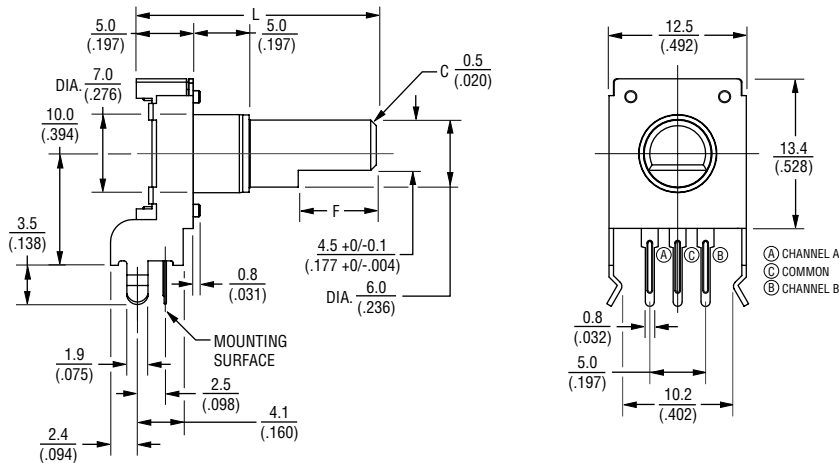
- Audio-visual equipment
- Consumer electric appliances
- Musical instrumentation
- Communications equipment

PEC12R - 12 mm Incremental Encoder

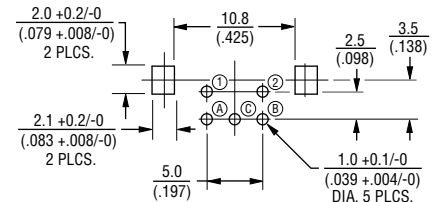
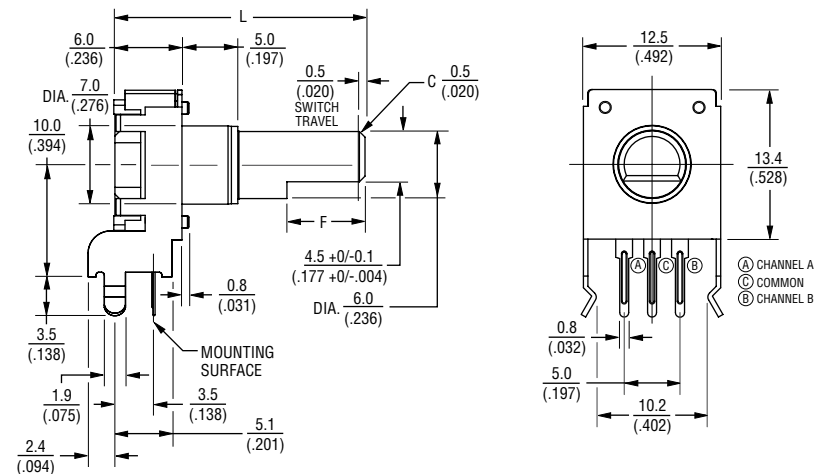
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Product Dimensions

PEC12R-2xxxF-Nxxxx (Vertical Mount - Radial PC Pin/No Bushing, No Switch)



PEC12R-2xxxF-Sxxxx (Vertical Mount - Radial PC Pin/No Bushing, Push Momentary Switch)



L	17.5 (.688)	20.0 (.787)	22.5 (.886)	25.0 (.984)	30.0 (1.181)
F	5.0 (.197)	7.0 (.276)	7.0 (.276)	12.0 (.472)	12.0 (.472)

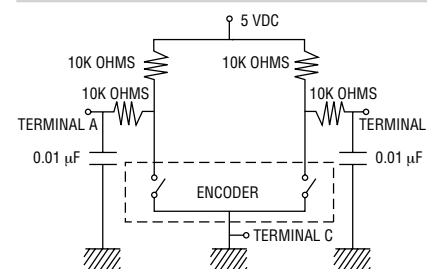
DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

TOLERANCE: $\frac{<10}{(<.400)} = \frac{\pm 0.3}{(\pm .012)}$
 $\frac{\geq 10}{(\geq .400)} = \frac{\pm 0.5}{(\pm .020)}$

Switch Circuit



Suggested Filter Circuit



Specifications are subject to change without notice.

Users should verify actual device performance in their specific applications.

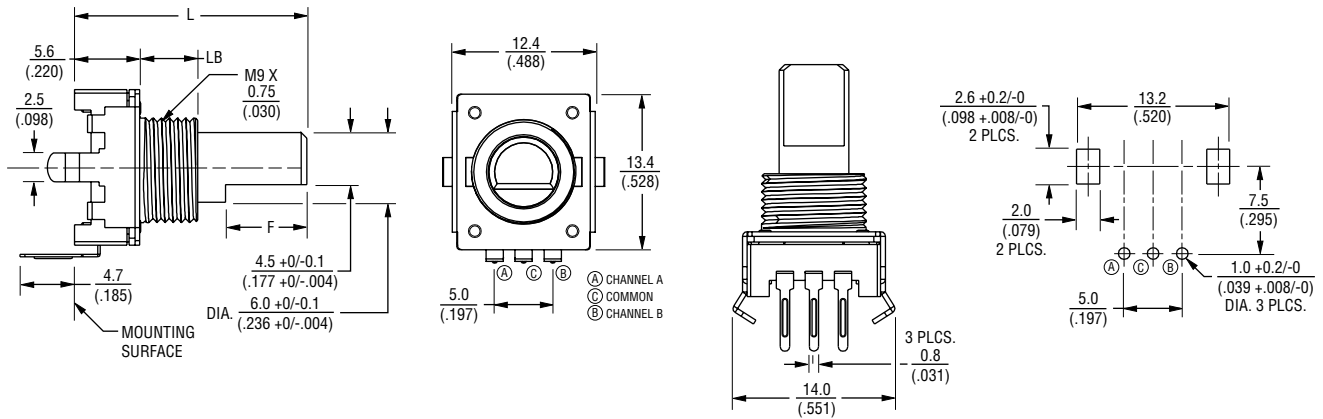
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PEC12R - 12 mm Incremental Encoder

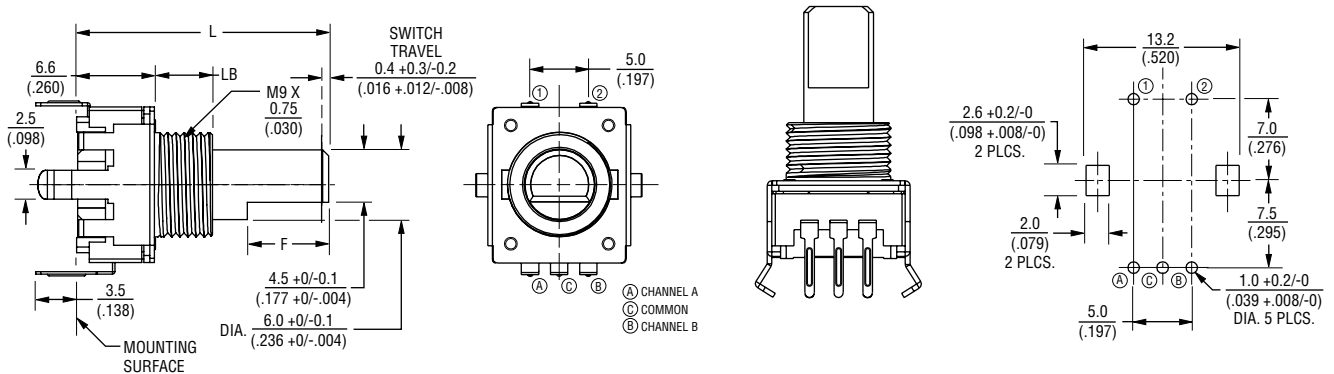
BOURNS®

Product Dimensions

PEC12R-3xxxF-Nxxxx (Horizontal Mount - Axial PC Pin/with Bushing, No Switch)



PEC12R-3xxxF-Sxxxx (Horizontal Mount - Axial PC Pin/with Bushing, Push Momentary Switch)



L	17.5 (.688)	20.0 (.787)	22.5 (.886)	25.0 (.984)	30.0 (1.181)
LB	5.0 (.197)	5.0 (.197)	7.0 (.276)	7.0 (.276)	7.0 (.276)
F	5.0 (.197)	7.0 (.276)	7.0 (.276)	12.0 (.472)	12.0 (.472)

DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

TOLERANCE: $\frac{<10}{(<.400)} = \pm 0.3$
 $\frac{\geq 10}{(\geq .400)} = \pm 0.5$

Specifications are subject to change without notice.

Users should verify actual device performance in their specific applications.

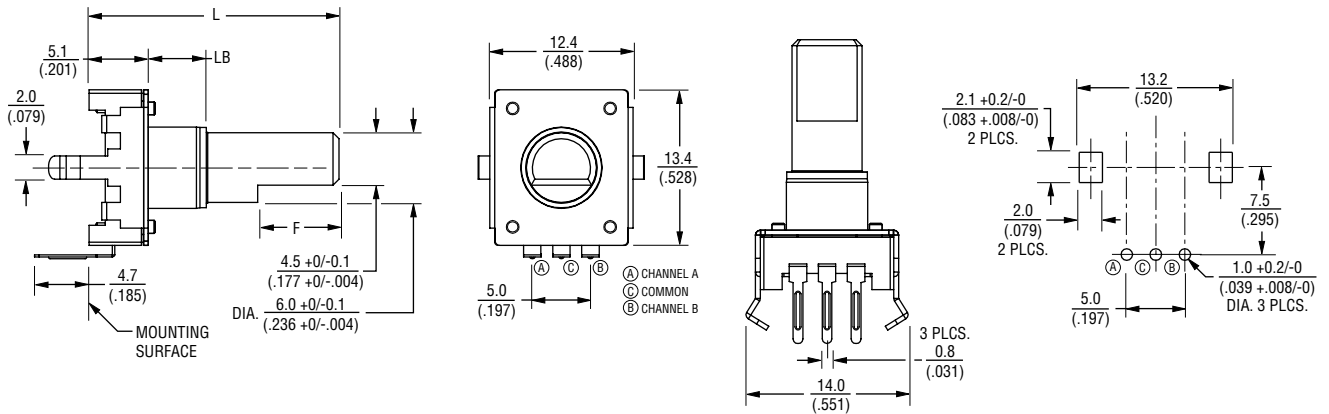
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PEC12R - 12 mm Incremental Encoder

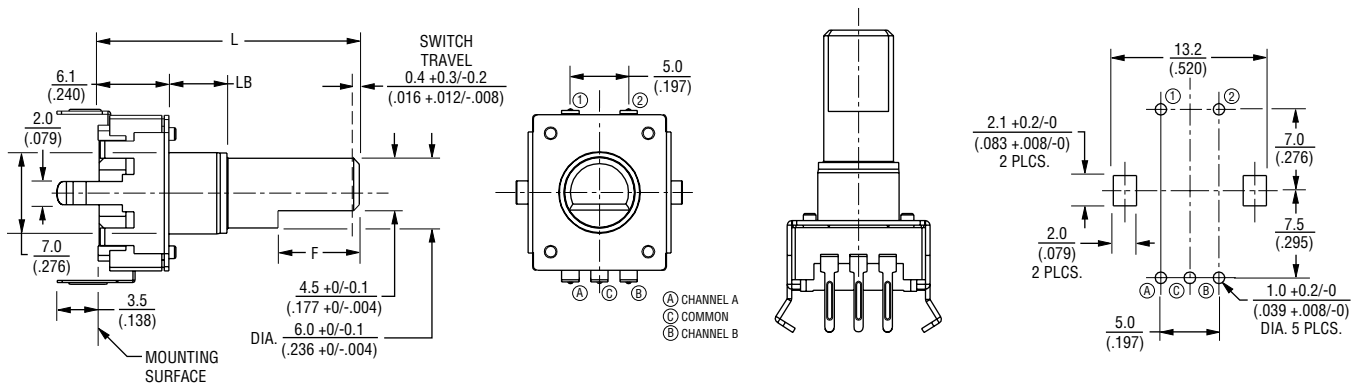
BOURNS®

Product Dimensions

PEC12R-4xxxF-Nxxxx (Horizontal Mount - Axial PC Pin/No Bushing, No Switch)



PEC12R-4xxxF-Sxxxx (Horizontal Mount - Axial PC Pin/No Bushing, Push Momentary Switch)



L	15.0 (.591)	17.5 (.688)	20.0 (.787)	22.5 (.886)	25.0 (.984)	30.0 (1.181)
LB	2.0 (.079)	5.0 (.197)	5.0 (.197)	5.0 (.197)	5.0 (.197)	5.0 (.197)
F	5.0 (.197)	5.0 (.197)	7.0 (.276)	7.0 (.276)	12.0 (.472)	12.0 (.472)

DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

TOLERANCE: $\frac{<10}{(<.400)} = \frac{\pm 0.3}{(\pm .012)}$
 $\frac{\geq 10}{(\geq .400)} = \frac{\pm 0.5}{(\pm .020)}$

REV. 07/21

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Users should verify actual device performance in their specific applications.

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